

Marketing Report - 2Market

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Business background

2Market is a international supermarket chain with two distribution channels: physical stores and online. The aim of the project is to understand consumer behaviour in order to maximise the effectiveness of online advertisement.
The insights will aim at answering the following questions:

1. What are the demographics of the 2Market consumer?
2. Which advertising channels are the most efficient at converting new purchases?
3. Which are the best selling products? Is there any relevant variations depending on the demographics of the consumer?

The final insights will be presented through the following deliverables:

- Tableau dashboard
- Final report
- Recorded presentation

Analytical approach

The first step to the analysis was cleaning the data to ensure the integrity of the results. Afterwards, the data was analysed in SQL and Excel in order to find the main trends, appropriate consumer segmentations and ultimately to cross-check the results obtained with the final dashboard in order

Data cleaning

The following modifications to the original data was implemented:

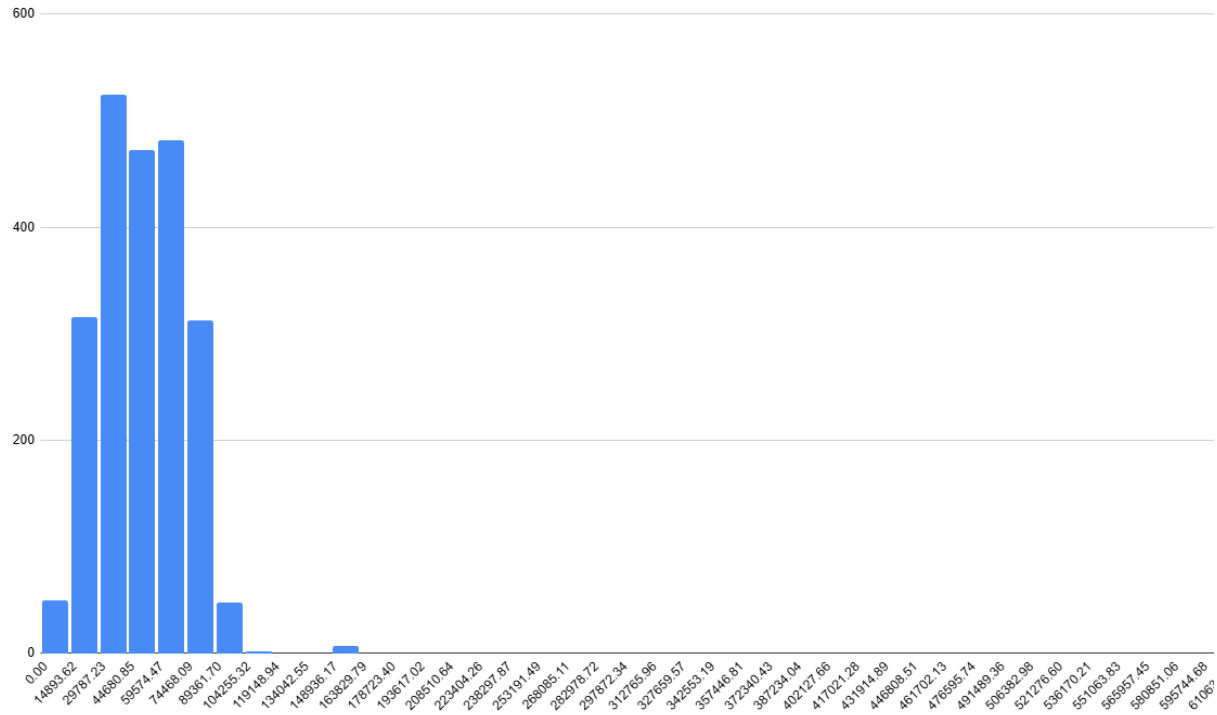
- Income and Dt_Consumer were converted from string to integer and date format respectively.
- In order to facilitate the data cleaning process and later the analysis, a new column containing the ages of each consumer was added.
- Consumers over the age of 100 were eliminated (see appendix for data collection year assessment).

ID	Year_Birth	Age
7829	1901	113
1150	1900	114
11004	1894	120

- Marital statuses set as “Alone”, “Absurd” and “YOLO” were replaced with null values as they are not consistent with the rest of the dataset.

ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teen
7734	1994	Graduation	Absurd	$79,244.00	0	0
92	1989	Graduation	Alone	$34,176.00	1	0
492	1974	PhD	YOLO	$48,432.00	0	1
11133	1974	PhD	YOLO	$48,432.00	0	1
7660	1974	PhD	Alone	$35,860.00	1	1
433	1959	Master	Alone	$61,331.00	1	1
4369	1958	Master	Absurd	$65,487.00	0	0

- Consumers with income above $102,000 were eliminated as they represent outliers compared to the rest of 2Market consumers, this was identified through the use of a histogram.



- 55 duplicates were eliminated to ensure unique records only are present.

7. duplicates check: in order to ensure no duplicates are present, the ID columnwas first checked for duplicates, which did not present any

ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer
8420	1971	2n Cycle	Married	$15,315.00	0	0	03/08/2013
10065	1971	2n Cycle	Married	$15,315.00	0	0	03/08/2013
10826	1960	Graduation	Married	$18,690.00	0	0	28/12/2012
10304	1960	Graduation	Married	$18,690.00	0	0	28/12/2012
2253	1991	Graduation	Married	$18,929.00	0	0	16/02/2013
2326	1991	Graduation	Married	$18,929.00	0	0	16/02/2013
9671	1969	Graduation	Married	$19,514.00	1	1	26/01/2014
8975	1969	Graduation	Married	$19,514.00	1	1	26/01/2014

- 2 records containing consumers that have had no purchases in-store or online were eliminated because not relevant for analysis.
- A new column containing boolean data aimed to inform whether a consumer is a parent or not was added, this solution was implemented to facilitate the analysis and remove unnecessary granularity.

Exploratory analysis on SQL and Excel

To begin the analysis the dataset needs to be imported into pgAdmin, using the metadata as guidance for data type specifications:

```sql
--Creation of tables --
CREATE TABLE marketing_data (
    "ID" INT,
    "Year_Birth" INT,
    "Education" VARCHAR (20),
    "Marital_Status" VARCHAR (20),
    "Income" INT,
    "Kidhome" INT,
    "Teenhome" INT,
    "Dt_Customer" DATE,
    "Recency" INT,
    "AmtLiq" INT,
    "AmtVege" INT,
    "AmtNonVeg" INT,
    "AmtPes" INT,
    "AmtChocolates" INT,
    "AmtComm" INT,
    "NumDeals" INT,
    "NumWebBuy" INT,
    "NumWalkinPur" INT,
    "NumVisits" INT,
    "Response" BOOLEAN,
    "Complain" BOOLEAN,
    "Country" VARCHAR (5),
    "Count_success" INT,
    "Age" INT,
    "Member_Since_Age" INT,
    "Parenthood" BOOLEAN) ;

CREATE TABLE ad_data (
    "ID" INT,
    "Bulkmail_ad" BOOLEAN,
    "Twitter_ad" BOOLEAN,
    "Instagram_ad" BOOLEAN,
    "Facebook_ad" BOOLEAN,
    "Brochure_ad" BOOLEAN);
```

After the data import, singular queries were ran, which allowed for the following decisions and insights:

- As the sample is disproportionately skewed towards Gen X consumers, age bracket segmentation was adjusted to group all the consumers under the age of 30 together, likewise for the less populated 60+ consumers; all the remaining ages were grouped at a 9 year intervals.
- As there are only 3 consumers from Montenegro, this country was excluded from the remaining analysis as it doesn't contain a sufficient sample size for a solid analysis.

Dashboard design and development

The following considerations were taken during the build stage of the dashboard:

- A left join was performed when importing the data, with marketing_data being the left join in order to avoid Ids deleted during the clean stage to appear in the ad_data table.

- The county codes (e.g. ES, AUS, CA) were converted to full country names (e.g. Spain, Australia, Canada) to allow stakeholders not familiar with the dataset to read the information displayed.
- Grid lines were removed from graphs to minimise visual weight.

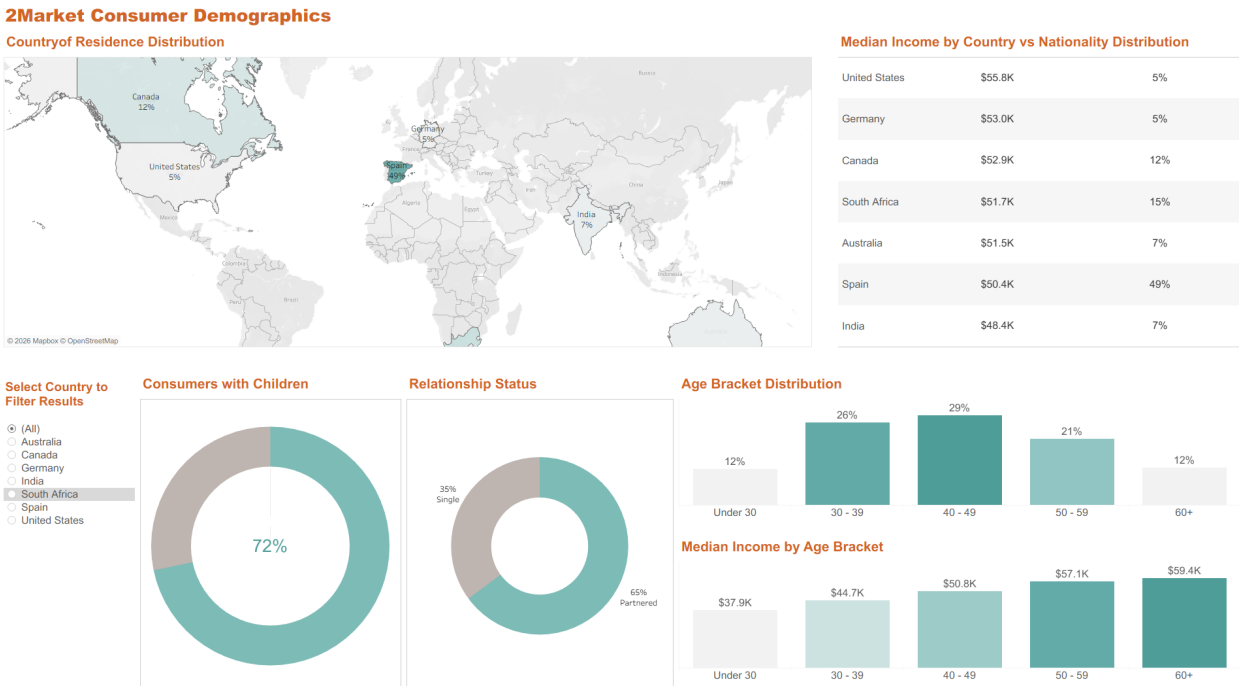


- The colour scheme selected was a simple 2-colour combination of complimentary colours (green and orange) to enhance visual design and create a clear distinction between the chart titles and data points.

- The size of the fonts create a clear hierarchy within the dashboard in order to guide the viewers across the screen and clearly identify information, with the smallest type being 12 pt and not less to keep it readable.



- Distribution of the elements across the dashboard was implemented with the general desktop layout in mind, as stakeholders will be most likely to access it through a company-issued laptop of desktop.



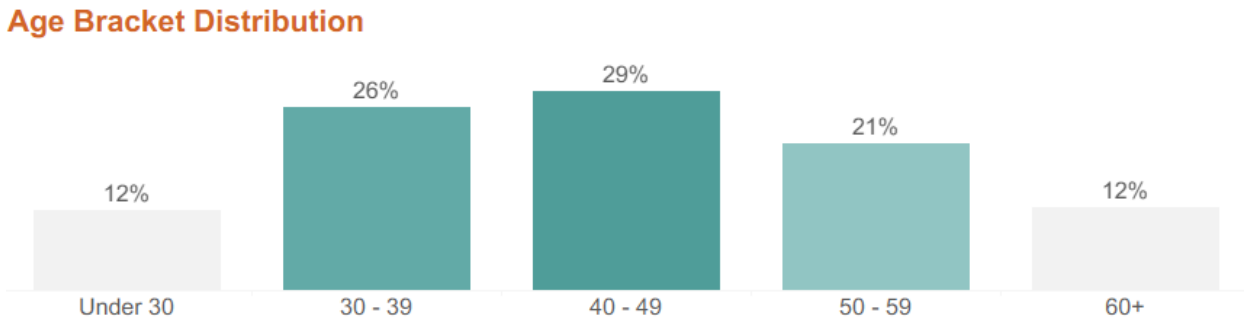
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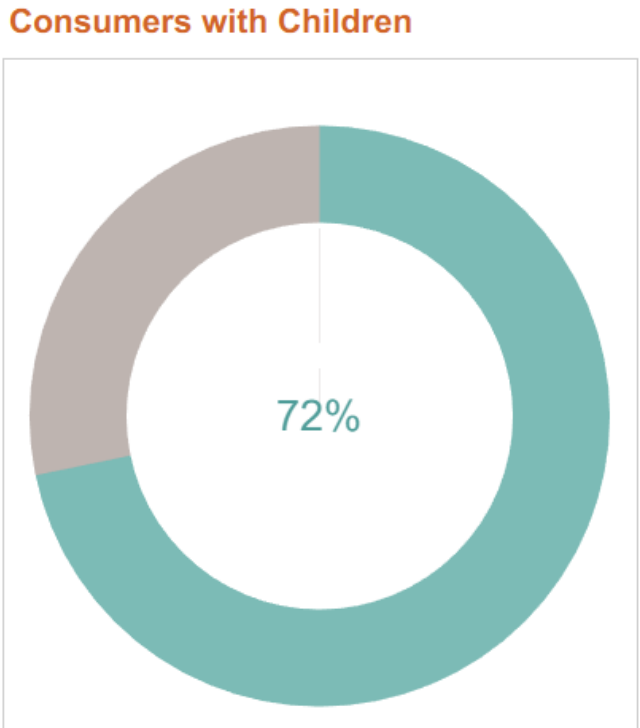
- The number of dashboards reflects the number of questions within the initial ask, to create a clear topic subdivision depending on the stakeholder needs.
- Since 2Market operates within several countries, regional nuances are important to capture and facilitate the reading for regional marketing managers, which will ultimately allow for market-relevant promotional initiatives; these considerations were translated in the addition of maps and interactive elements to filter data by the country consumers reside in.

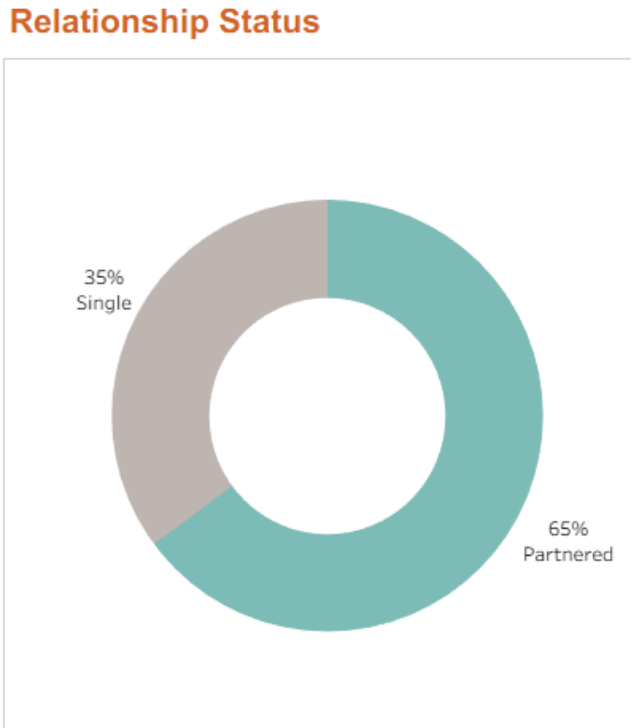
  This approach allows for a funnel-like view of the dataset, where the user can navigate starting from the top-line information into higher detail.
- Decimal numbers were eliminated in order to avoid over-complicating the analysis with too much granularity.
- As tableau does not allow to import currencies in a number format, raw data was edited to eliminate the dollar sign. Income and spend information was formatted during the build stage, with bigger numbers (like income) being displayed in thousands to aid market-by-market comparisons.

Patterns, trends and insights

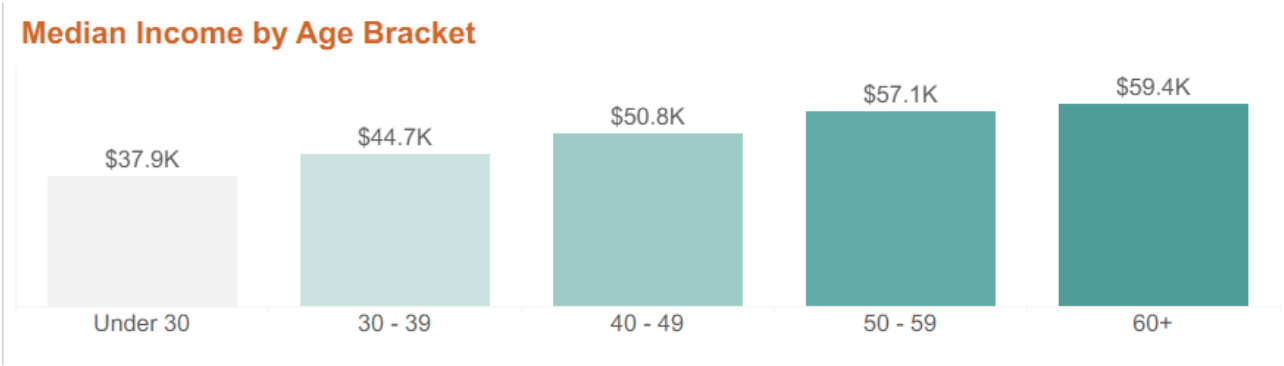
- The 2Market consumer base is highly polarised towards Spain, which holds almost half of all purchasers; however, it is a market with one of the lowest median incomes, meaning that expansion towards markets with higher disposable income is likely to raise 2Market revenue.
- 2Market presents a mature demographic, with the great majority of consumers having a partner and offspring, meaning that a family-oriented marketing narrative is recommendable.



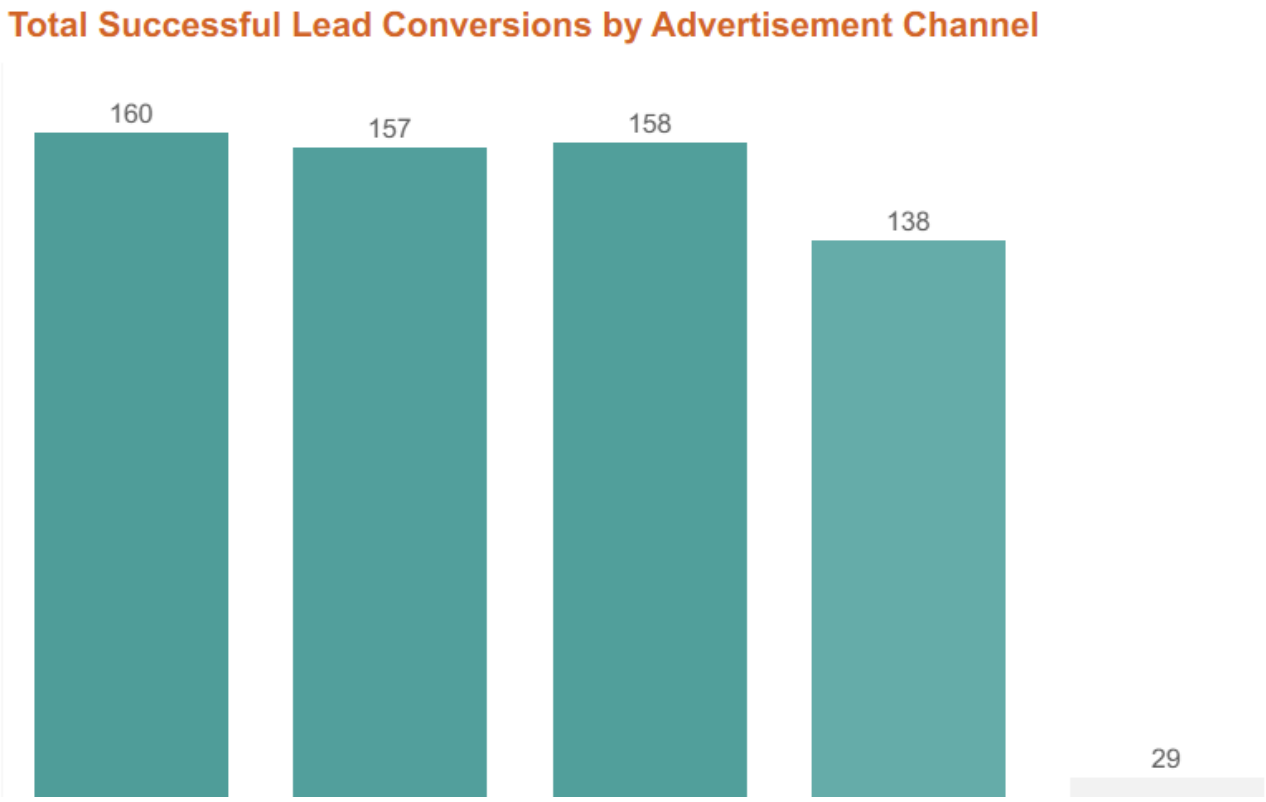


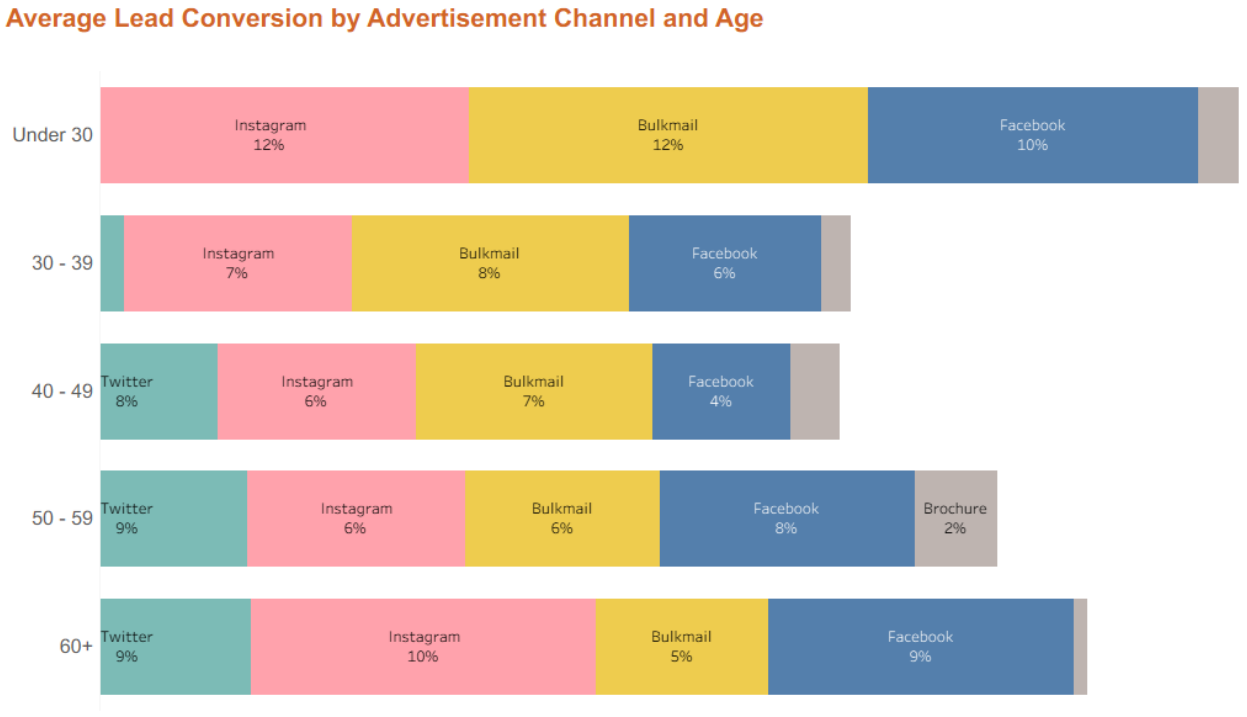


- Pushing more premium products towards older demographics is recommendable, as the older the consumer is, the higher their income is likely to be.



- De prioritising brochures in favour of Twitter, Instagram and Bulk-mail ads would be best, although the age of the consumer is and important factor to consider; in fact, successful lead conversions for the under-30 consumers is more likely to be from Instagram, Bulkmails and Facebook, moreover, they are more likely to engage with ads in general, which highlights the opportunity for 2Market to broaden its consumer base with this demographic.





- Most of 2Market sales come from alcohol and meat sales, where the older the consumer, they more they are likely to purchase alcohol.
- Consumers with kids/teens spend significantly less on alcohol

Appendix – Analytical approach

Before the start of the analysis, it is important to ensure data integrity, which starts with the examination and cleaning of the raw data provided. The following steps were undertaken for the marketing_data:

- the **type of the data** was checked against the metadata, as a consequence, the Income column got converted to numerical values from string Dt_Customer was converted to a date format
- a **sense check of consumer ages**: consumers over the age of 110 were eliminated, along with a record featuring a married consumer of the age of 17, married and member since the age of 15 and purchasing alcohol. In 2014 Australia (where the consumer resides) marriage under the age of 18 is only permitted through a court permit under exceptional circumstances, the purchase of alcohol is not legal for those under the age of 18 and it is highly unlikely for a 15 year old to join the supermarket chain, which suggests that an error during the data collection stage is likely to have occurred.
    - To establish the year of the data collection, the Recency and Dt_Customer column were utilised: with a Recency of 0 and last year a customer joined was 2014. Historical data suggests that there is globally at least one new consumer joining every year, therefore, it is most likely that the year of collection was 2014. A new “Age” column was added to the dataset in order to aid with the analysis and potential segmentation of age brackets.
- **String content check**: columns containing string values were investigated. Marital Status column had inconsistent values, "Alone" could have been replaced with "Widowed", "Divorced" or "Single" status but without further information, it is not possible to assess which of these options is most suited, therefore, the values were replaced with null. Other string columns did not present anomalies.
- String spacing check: in order no extra spacing was included within the strings (which could impede the queries from working) a check was performed comparing the results of the =TRIM formula against the original dataset, no discrepancy was found.
- **Outlier detection**: after the elimination of the outliers in the Age column, the Income column was checked through the use of a histogram, which led to the elimination of all records containing an income above $102,000.
- **Elimination of duplicates**: in order to ensure no duplicates are present, the ID column was first checked for duplicates, which did not present any. To ensure new IDs were not generated automatically for existing consumers, the Year_Birth, Income, Dt_Customer and Country column were checked. In order to find the records with these column in common, a helper column (which was later deleted) was created: it creates a merge of the contents of the columns analysed and those which were counted more than once represent duplicates which were eliminated.
- **Elimination of irrelevant sample**: as the aim of the analysis is to find insights around 2Market consumers, it is necessary that every person in the dataset has made at least one purchase with the store. In order to ensure the sample passes this criteria, the columns NumWalkinPur, NumWebBuy and NumDeals were analyse, as they refer to numbers of purchases.

Data analysis in SQL & excel approach

Initial observations were based on SQL queries, which were also later utilised in order to cross-

check the final results represented in the dashboard for correctness.
In addition to pulling data through SQL, it was additionally worked with in excel to allow quick formatting and calculations.
To begin with, the tables designated to host the datasets were created following the metadata file to ensure the right data type is selected:

```
CREATE TABLE marketing_data (
        "ID" INT,
        "Year_Birth" INT,
        "Education" VARCHAR (20),
        "Marital_Status" VARCHAR (20),
        "Income" INT,
        "Kidhome" INT,
        "Teenhome" INT,
        "Dt_Customer" DATE,
        "Recency" INT,
        "AmtLiq" INT,
        "AmtVege" INT,
        "AmtNonVeg" INT,
        "AmtPes" INT,
        "AmtChocolates" INT,
        "AmtComm" INT,
        "NumDeals" INT,
        "NumWebBuy" INT,
        "NumWalkinPur" INT,
        "NumVisits" INT,
        "Response" BOOLEAN,
        "Complain" BOOLEAN,
        "Country" VARCHAR (5),
        "Count_success" INT,
        "Age" INT,
        "Member_Since_Age" INT) ;

CREATE TABLE ad_data (
        "ID" INT,
        "Bulkmail_ad" BOOLEAN,
        "Twitter_ad" BOOLEAN,
        "Instagram_ad" BOOLEAN,
        "Facebook_ad" BOOLEAN,
        "Brochure_ad" BOOLEAN);
```

Answering question 1: what are the demographics of the 2Market consumer?

Age bracket distribution: After importing the data, the following query aimed to identify how many consumers belong to each age was ran:

```
SELECT "Age", COUNT("Age")
FROM marketing_data
GROUP BY "Age"
ORDER BY "Age" ASC;
```

The resulting data was then imported into excel for further analysis where the following

observations were made:

- 2Market enjoys a rather mature consumer base, with the highest incidence of consumers in their late 30s and 40s, with the average consumer age being 38
- Splitting consumers into generational brackets would not have been the best approach as it would over-represent the Silent Generation (with only 15 consumers belonging to it) and would have risked to lose detail for the Gen X consumers (with 1026 consumers):

Generation	Count
Millennials	407
Gen X	1026
Boomers	696
Silent Generation	15

Therefore, the following segmentation was decided for the analysis:

Age bracket	Count	%
Under 30	248	12%
30 - 39	567	26%
40 - 49	618	29%
50 - 59	453	21%
60+	258	12%

- The initial insight deducted is the fact that 30-49 are crucial age brackets for 2Market to cater to, as they represent more than half of its consumer base (55%), with a special focus to be given to the consumers aged 40 – 49.

Country of origin:

SQL query:

```
SELECT "Country", COUNT ("Country")
FROM marketing_data
GROUP BY "Country"
ORDER BY COUNT ("Country") DESC;
```

Output:

Country label	Count	%
Spain	1041	49%
South Africa	331	15%
Canada	261	12%
Australia	145	7%
India	143	7%
Germany	116	5%
United States	105	5%
Montenegro	3	0%

Observations:

Spain represents by far the market with the broadest consumer base, The remaining consumers are rather diversified, as the company is present across several continents, with South Africa and Canada being particularly relevant countries after Spain.

Only 3 consumers are resident in Montenegro, which significantly under-indexes the rest of the dataset and is not a sufficient sample to identify actionable recommendations, meaning that this market is to be excluded from the rest of the analysis.

Parenthood:
Addition of extra columns for parenthood boolean: to simplify analysis, as the dataset does not contain products aimed directly at teens or infants/kids, reducing the granularity of the information would be helpful. The original dataset contains the number of kids or teens each consumer has, however, just knowing whether the consumers are purchasing for multiple members within a household would help make clearer insights.
Since a new column was added in the dataset, the SQL query for the creation of the table was amended to include the new Parenthood column. In order to keep documentation up to date, the 2Market team will be alerted of the addition, so the nature of the data contained can be inputted in the metadata file.

After querying the new attribute through the following SQL syntax:

```
SELECT "Parenthood", COUNT ("Parenthood")
FROM marketing_data
GROUP BY "Parenthood";
```

Output:

Parenthood	count	
FALSE	606	28%
TRUE	1537	72%

Observations:
Most of consumers are parents, probably due to the mature nature of the 2Market audience; to verify this hypothesis, a query aimed at identifying the parenthood status of consumers by age bracket:

```
SELECT
   CASE
      WHEN "Age" < 30 THEN 'Under 30'
      WHEN "Age" BETWEEN 30 AND 39 THEN '30 - 39'
      WHEN "Age" BETWEEN 40 AND 49 THEN '40 - 49'
      WHEN "Age" BETWEEN 50 AND 59 THEN '50 - 59'
      ELSE '60+'
   END AS Age_Bracket,
   COUNT(*) AS Total_Consumers,
   COUNT(*) FILTER (WHERE "Parenthood" = TRUE) AS Total_Parents
FROM marketing_data
GROUP BY Age_Bracket
ORDER BY Age_Bracket;
```

In order to accomplish this, the CASE statement was utilised to create temporary age brackets for the query (which saves time creating a custom column for it), additionally, aliases were utilised to facilitate the query syntax.

Output:

age_bracket	total_consumers	total_parents	**% parents vs total age bracket**	**parents distribution among all consumers**	**total age bracket distribution**
Under 30	248	131	53%	9%	12%
30 - 39	567	443	78%	29%	26%
40 - 49	618	496	80%	32%	29%
50 - 59	453	321	71%	21%	21%
60+	257	146	57%	9%	12%

Observations:
The youngest and the oldest consumers are the least likely to be parents, yet, at least half of the total consumer base are parents, meaning that family-oriented marketing initiatives are likely to be particularly relevant to cater to consumers' lifestyle, especially consumers aged 30 – 49.

Average consume income:
Understanding the purchasing power dynamics is important to calibrate pricing and understand where to invest resources for higher capital return. As incomes can variate based on the country consumers reside in (as minimum wages and cost of living may differ), a country-by-country view is most beneficial to understand the full picture.

To extract this information via SQL the following syntax was applied:

```
SELECT "Country", AVG("Income") AS Average_Income
FROM marketing_data
GROUP BY "Country"
ORDER BY Average_Income DESC;
```

Output:
The final results were compared to the 2Market consumer density across countries

Country	average_income	population distribution
ME	$57,680	0%
GER	$52,951	5%
CA	$52,862	12%
SA	$52,792	15%
US	$52,219	5%
AUS	$52,032	7%
SP	$51,183	49%
IND	$48,434	7%

Observations:
Although Spain represents the main 2Market consumer base, it is one of the countries with the lowest incomes, meaning that 2Market could potentially benefit from expanding into countries where its presence is less prevalent, like Montenegro, Germany or Canada.

Answering question 2: which advertising channels are the most efficient at converting new purchases?

To approach this task, metadata was consulted. Marketing_data dataset contains 2 fields which could be useful for the analysis: Count_success and Response, which can indicate how likely a consumer or a set of consumers can be to interact and be influenced by advertisement channels.

However, Ad_data will be the main source of information to use, as it already contains the information which will be the most helpful to answer the question, as it contains the success lead conversions from each channel.

Total lead conversions by channel

To begin with, it would be interesting to observe which channels contain the biggest number of successful lead conversions through the following query:

```
SELECT
        SUM ("Bulkmail_ad"::INT) AS Bulkmail,
        SUM("Twitter_ad"::INT) AS Twitter,
        SUM("Instagram_ad"::INT) AS Instagram,
        SUM("Facebook_ad"::INT) AS Facebook,
        SUM("Brochure_ad"::INT) AS Brochure
FROM ad_data a
RIGHT JOIN marketing_data m
ON a."ID"=m."ID";
```

Note: As each column in a table, the ::INT allow to convert each field into an integer, which allows to use SUM as opposed to counting TRUE values only, which simplifies the query.

A right join with marketing_data was added because the ad_data table contains more records than marketing_data, meaning that consumers which were excluded during the data cleaning stage could appear if ad_data was analysed by itself, potentially skewing the results.

Output:

		share of success rate
bulk-mail	158	24.6%
twitter	160	24.9%
Instagram	158	24.6%
Facebook	138	21.5%
brochure	29	4.5%

Observations:

All social media advertisement channels share an almost equal portion of the total successful lead conversions (with Facebook performing slightly less); in the meanwhile, brochures tend to be significantly less effective.

The lower brochure effectiveness could be due to to the fact that most consumers shop online and are therefore less likely to interact with physical brochures, which are typically kept in-store.

This hypothesis was tested using the following query:

```
SELECT
        SUM("NumWebBuy") AS Total_Web,
        SUM("NumWalkinPur") AS Total_Walkin
```

FROM marketing_data;

Output:

total_web	total_walkin
8800	12489
41%	59%

Observations

The hypothesis was disproven, as most visits happen in-store, signifying that 2Market should either de prioritise brochures or potentially make them more appealing to consumers.

Answering question 3: Which are the best selling products? Is there any relevant variations depending on the demographics of the consumer?

Based on the nature of the dataset, it is possible to gain information around the amount spent on several food items, and therefore identify the categories producing the most revenue.

To extract the sales per category the following query was run:

```
SELECT
        SUM("AmtLiq") AS Alcohol,
        SUM("AmtVege") AS Vegetables,
        SUM("AmtNonVeg") AS Meat,
        SUM("AmtPes") AS Fish,
        SUM("AmtChocolates") AS Chocolate,
        SUM("AmtComm") AS Commodities
FROM marketing_data;
```

Output:

product	**sales amount**	**sales share**
alcohol	657,410	50.4%
meat	356,207	27.3%
commodities	94,455	7.2%
fish	80,756	6.2%
chocolate	58,114	4.5%
vegetables	56,472	4.3%
total sales	**1,303,414**	

Observations:

The sales present themselves highly polarised towards the top 2 products, which represent 77.8% of all the sales registered.

It would be interesting to understand whether age influences the type of product purchased (e.g. are older consumers more likely to moderate their alcohol consumption?).

The CASE applied earlier was utilised to break down consumers by age:

```
SELECT
        CASE
    WHEN "Age" < 30 THEN 'Under 30'
    WHEN "Age" BETWEEN 30 AND 39 THEN '30 - 39'
    WHEN "Age" BETWEEN 40 AND 49 THEN '40 - 49'
    WHEN "Age" BETWEEN 50 AND 59 THEN '50 - 59'
```

```
        ELSE '60+'
            END AS Age_Bracket,
        SUM("AmtLiq") AS Alcohol,
        SUM("AmtVege") AS Vegetables,
        SUM("AmtNonVeg") AS Meat,
        SUM("AmtPes") AS Fish,
        SUM("AmtChocolates") AS Chocolate,
        SUM("AmtComm") AS Commodities
FROM marketing_data
GROUP BY Age_Bracket
ORDER BY Age_Bracket;
```

Output:

age_bracket	alcohol	vegetables	meat	fish	chocolate	commodities
Under 30	60,220	7,323	47,293	9,560	7,458	10,822
30 - 39	142,684	13,871	80,254	20,583	14,561	22,056
40 - 49	184,600	15,329	95,328	20,051	15,530	26,270
50 - 59	167,588	12,790	81,159	19,266	13,172	21,850
60+	102,318	7,159	52,173	11,296	7,393	13,457

Observations:

The results returned by the query are simply skewed towards the most populated consumer groups, therefore it is not easy to identify trends for different consumers.

To get around this issue, the average amount spend by each consumer was calculated by dividing each amount by the total number of consumers within each age bracket in excel. An additional column for the total average spend by consumer type was created to identify who tends to generally spend the most.

Output:

age_bracket	alcohol	vegetables	meat	fish	chocolate	commodities	total
Under 30	243	30	191	39	30	44	575
30 - 39	252	24	142	36	26	39	519
40 - 49	298	25	154	32	25	42	577
50 - 59	370	28	179	43	29	48	697
60+	397	28	202	44	29	52	751

Observations:

- The oldest consumers tend to spend the most, with a significant peak for everyone aged 50+
- Unlike the initial expectations, older consumers tend to actually spend the most on alcohol, whilst the opposite is true for the under 30.
- Little variation in the average vegetable sales can be observed, probably due to the necessity of the product

The higher spend from the oldest consumer might be due to higher incomes, which can be verified by slightly editing the previous query:

```
SELECT
        CASE
      WHEN "Age" < 30 THEN 'Under 30'
      WHEN "Age" BETWEEN 30 AND 39 THEN '30 - 39'
      WHEN "Age" BETWEEN 40 AND 49 THEN '40 - 49'
      WHEN "Age" BETWEEN 50 AND 59 THEN '50 - 59'
      ELSE '60+'
          END AS Age_Bracket,
        AVG("Income") AS Total_Income
FROM marketing_data
GROUP BY Age_Bracket
ORDER BY Age_Bracket;
```

Output:

age_bracket	total_income
Under 30	45,421.69
30 - 39	47,924.53
40 - 49	51,374.76
50 - 59	56,354.47
60+	58,374.22

Observations:

The result of the query confirms the hypothesis: the older the consumer, the more they earn. Considering the relationship between amount spent in the supermarket and age/income, a breakdown of incomes by age bracket might be an interesting metric to ad to the dashboard.